Replicating portfolios

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The need of replicating portfolios in insurance: Solvency 2 regulation

- In the current regulatory framework (Solvency II)
- insurance companies have to hold enough capital
- to survive a bicentenary event (we talk about capital requirement).

- More precisely, the probability of insolvency of an insurance company
- in a one year horizon, has to be lower than 0.5%.

- To meet the regulation, an insurance company
- choosing to rely on its internal model
  - insurance companies can also choose to directly apply the standard formula
  - provided by the regulator
- then has to compute the distribution of its profit and losses over one year.

The need of replicating portfolios in insurance: Liability projection

- Such a computation, require the projection of both asset and liabilities
- in a large variety of scenarios.
  - [Vidal et Daul, 2009, Devineau et Chauvigny, 2010]

- Whereas projecting (or simulating) the evolution of insurance company's asset
- over one year, correspond to applying standard methods
  - as insurers mostly hold standard financial assets in their portfolio;
- projecting their liability is more complex (and computational costly).

- Insurers' liability indeed corresponds to insurance contracts
- and simulating its evolution comes to simulating the behavior of policyholders
- both in terms of claims and in terms of contracting choices.

The definition of replicating portfolios in insurance

- One way to ease this projection and reduce computations
- is to replicate the behavior of insurer liability
- with a portfolio of standard financial assets.

- The aim is then to find the asset portfolio that fits the best the evolution
- of liability, that is of insurance contracts portfolio.

- As this is ultimately used to compute the distribution of P&L over one year
- this fits as not only to be good in "standard" cases
- but also on the tails of the distribution.

Scenarios and simulations

- To do so, insurers rely on "nested simulations":
  - drawing first an "economic scenario", that is a set of economic indicators (inflation, interest rate, stock returns,...) called risk factor
  - and performing for each of these scenarios Monte Carlo simulations on the market values of assets and liabilities
    - side note: some replicating portfolio models use cash flow instead of market values
  - To finally compute the distribution of P&L.

- In the following, we denote by
  - $k = 1, ..., K$ the economic scenarios and by
  - $s = 1, ..., S$ the Monte Carlo simulations

- The aim of the replicating portfolio is to save on Monte Carlo simulations on the liability side

Our problem: building and calibrating a replicating portfolio

Denoting by

- $Y_{k,s}$ the market value of insurer liability in simulation s of scenarios k
- $X_{i,k,s}$ the market value of asset i in simulation s of scenarios k

Our aim is to find

- a set $\mathcal{I}$ of assets (selection stage)
- a position β_i on each of these assets (estimation stage)

To approximate $(Y_{k,s})_{k,s}$ by:

$$\sum_{i \in \mathcal{I}} \beta_i * (X_{i,k,s})_{k,s}$$

Data

- For 5212 assets (X)
- and one group of contracts (Y)

1. the market values among the 4000 simulations for the "real-world" (unstressed) scenario, labeled scenario 1:

$$\left(Y_{1,s}, \left(X_{i,1,s}\right)_{i=1}^{5212}\right)_{s=1}^{4000}$$

2. the average market value among the 4000 simulations for the other 45 stressed scenarios (change in key "economic" variables):

$$\left(\overline{Y}_k, \left(\overline{X}_{i,k}\right)_{i=1}^{5212}\right)_{k=1}^{56}$$

The problem

We want to find

- the subset of assets / instruments X_i
- and for each a position β_i

that

- is not too large (to limit overfitting)
- replicates well the behavior of liability Y on average
- provides good enough fit in each stressed scenario

DEVINEAU, L. et CHAUVIGNY, M. (2010).
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VIDAL, E. G. et DAUL, S. (2009).
Replication of Insurance Liabilities.
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